

EduGenie Project

Topic 2 — explanation_module.py — Concept Explanation Logic

```
from config import ENABLE_LOCAL_EXPLAINER, LOCAL_EXPLAINER_MODEL
from gemini_client import generate_text

_local_pipeline = None

def _get_local_pipeline():
    global _local_pipeline
    if _local_pipeline is None:
        from transformers import pipeline
        _local_pipeline = pipeline(
            "text2text-generation",
            model=LOCAL_EXPLAINER_MODEL,
        )
    return _local_pipeline

def _local_explain(topic: str) -> str:
    pipe = _get_local_pipeline()
    prompt = (
        "Explain this educational topic for a beginner in simple language. "
        "Use short paragraphs and examples when useful. Topic: " + topic
    )
    result = pipe(prompt, max_new_tokens=220, do_sample=False)
    return result[0]["generated_text"].strip()

def explain_topic(topic: str) -> str:
    if ENABLE_LOCAL_EXPLAINER:
        try:
            return _local_explain(topic)
        except Exception:
            # Keep the application usable if the optional local model is unavailable.
            pass

    prompt = f"""
    Explain the following educational topic to a beginner.
    Structure the answer as:
    1. Simple definition
    2. How it works / key idea
    3. Small example
    4. One-line takeaway

    Topic:
    {topic}
    """
    return generate_text(prompt)
```